# Java fra scratch

# Program

- 08:30 Lidt om læring, AI og fremtiden som programmør
- 09:00 Hvad er Java
- 09:30 Installering af Java og IntelliJ
- 10:00 Pause
- 10:15 Første program
- 11:15 Opsamling

# Hardware og software

Computeren er en **standardmaskine**, der kan udføre mange **forskellige opgaver**

# Hardware

- **CPUen** **beregner** og udfører instrukser
- **RAM** er **midlertidig** arbejdshukommelse
- **Harddisk** er **permanent** hukommelse - filer

# Software

- **Operativsystem** styrer computeren
  - fx Windows, MacOS, Android, iOS
- **Drivere** får computeren til at kommunikere med hardware
  - fx keyboard, printer, skærm
- **Applikationer** er det, vi bruger computeren til
  - fx Word, Photoshop, browsere, spil

# Programmeringssprog


مرحبا
( marHaba )
ARABIC
Здравствуйте
( zdrastvooite )
RUSSIAN
Hola
( ola )
SPANISH
Hallo
( halo )
GERMAN
Bonjour
( boñ-zhoor )
FRENCH
Ciao
( chow )
ITALIAN
Olá
( o-laa )
PORTUGUESE
नमस्ते
( namastey )
HINDI

# Hello World in 30 different languages

**C**
```c
#include

int main(void)
{
    puts("Hello, world!");
}
```

**Matlab**
```matlab
disp('Hello, world!')
```

**Pascal**
```pascal
WriteLn('Hello, world!');
```

**Go**
```go
println('Hello, world!');
```

**F#**
```fsharp
printfn "Hello World"
```

**Lisp**
```lisp
(print "Hello world")
```

**C#**
```csharp
Console.WriteLine("Hello, world!");
```

**Ruby**
```ruby
puts "Hello World!"
```

**Java**
```java
System.out.println("Hello World!");
```

**JavaScript**
```javascript
console.log 'Hello, world!'
```

**C++**
```cpp
#include

int main()
{
    std::cout << "Hello, world!
";
    return 0;
}
```

**CoffeeScript**
```coffeescript
console.log 'Hello, world!'
```

**Python**
```python
print('Hello, world!')
```

**PHP**
```php
echo "Hello World!";
```

**Algol**
```
BEGIN DISPLAY("HELLO WORLD!") END.
```

**Delphi**
```delphi
program HelloWorld;
begin
  Writeln('Hello, world!');
end.
```

**Assembly**
```asm
    global  _main
    extern  _printf

    section .text
_main:
    push    message
    call    _printf
    add     esp, 4
    ret
message:
    db  'Hello, World', 10, 0
```

**Pascal**
```pascal
program HelloWorld(output);
begin
  Write('Hello, world!')
end.
```

**Perl**
```perl
print "Hello, World!\n";
```

**Tcl**
```tcl
puts "Hello World!"
```

**Cobol**
```cobol
       IDENTIFICATION DIVISION.
       PROGRAM-ID. hello-world.
       PROCEDURE DIVISION.
           DISPLAY "Hello, world!"
           .
```

**Dart**
```dart
main() {
  print('Hello World!');
}
```

**Kotlin**
```kotlin
fun main(args: Array<String>) {
    println("Hello World!")
}
```

**TypeScript**
```typescript
console.log 'Hello, world!'
```

**Scala**
```scala
object HelloWorld extends App {
   println("Hello, World!")
 }
```

**Haskell**
```haskell
module Main where

main :: IO ()
main = putStrLn "Hello, World!"
```

**R**
```r
cat("Hello world\n")
```

**Swift**
```swift
println('Hello, world!');
```

**HTML**
```html
Hello world
```

**Fortran**
```fortran
program helloworld
     print *, "Hello world!"
end program helloworld
```

| May 2025 | May 2024 | Change | Programming Language | Ratings |
|---|---|---|---|---|
| 1 | 1 | | Python | 25.35% |
| 2 | 3 | | C++ | 9.94% |
| 3 | 2 | | C | 9.71% |
| **4** | **4** | | **Java** | **9.31%** |
| 5 | 5 | | C# | 4.22% |
| 6 | 6 | | JavaScript | 3.68% |
| 7 | 8 | | Go | 2.70% |
| 8 | 7 | | Visual Basic | 2.62% |
| 9 | 11 | | Delphi/Object Pascal | 2.29% |

Java er et **general purpose programmeringssprog**


Syntaks

**Syntax** - regler for rækkefølge af ord *"spiser en sandwich jeg"*

**Udsagnsled** + **grundled** + **genstandsled**

*eller*

*"jeg spiser en sandwich"*

**grundled** + **udsagnsled** + **genstandsled**

**Korrekt**

```
int age = 18;
```

**Forkert**

```
age = 18 int;
```

```
int = 18 age;
```

```
INT age = 18;
```

**Korrekt**

```
System.out.println("Hello, World!");
```

**Forkert**

```
System.out.println Hello, World!;
```

```
System.out.println(Hello, World!");
```

```
System.out println("Hello, World!");
```

**Korrekt**

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

**Forkert**

```java
public class HelloWorld {}
    public static void main(String[] args) {
        System.out.println("Hello, World!");
```

```java
public class HelloWorld {
    public static void (String[] args) {
        System.out.println("Hello, World!");
    }
}
```

```java
public HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

# Første program

1. Lav en fil, der hedder `HelloWorld.java`
2. Skriv følgende kode i filen og gem den:

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

3. Compile med

```
javac HelloWorld.java
```

4. Kør programmet med

```
java HelloWorld
```

5. Hvad sker der?

# Kompilering

Compileren `javac` oversætter kildekode `HelloWorld.java` til bytecode `HelloWorld.class`

```
javac HelloWorld.java
```

# Java bytecode

er en binær fil, der ender på `.class`

# Whitespace

- mellemrum, tabulatorer og linjeskift
- **ikke** vigtigt for computeren

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

er **korrekt** syntaks

```java
public class HelloWorld {
public static void main(String[] args) {
System.out.println("Hello, World!");
}
}
```

er **også korrekt** syntaks

```
public class HelloWorld { public static void main(String[] args) { System.out.println("Hello, World!"
```

er **ogsÃ¥ korrekt**

Hvorfor synes I vi skal bruge **indrykning**?

Tre ting du tager med fra fra i dag?